SUBSTRING SEARCH

1) BRUTE FORCE

pattern - SUB

i	j	0 S	1 U	2 R	3 S	4 N	5 G	6 G	7 S	8 S	9 U	10 B	S	U
0	2	S ✓	U ✓	B ✗										
1	0		S ✗											
2	0			S ✗	S ✓	X								
3	1					U ✗	S ✗							
4	0							S ✗						
5	0								S ✗					
6	0									S ✓	U ✗			
7	1										S	U	B ✓	
8	3													

i - index

j - nr. i karaktereve qe i kemi gjet

↳ return 8;

- substring

Scanned with CamScanner

2. BOYER-MOORE

Pattern: BEES

i	j	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
		E	S	A	B	E	C	S	E	B	B	E	E	S	S	A
0	3	B	E	E	S											
3	4				B	E	E	S								
									B	E	E	S				
7	2										B	E	E	S		
9																

return i=9

Scanned with CamScanner

(2) pattern: NEEDLE

		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
i	j	F	I	N	D	I	N	A	H	A	Y	S	T	A	C	K	N	E	E	D	L	E
0	5	N	E	E	D	L	E	✗														
5	6						N	E	E	D	L	E	✗									
11	4												N	E	E	D	L	✗ E	✓ E	✓ D	✓ L	✓ E
15	0																	N	E	E	D	L	E

→ return i=15

Scanned with CamScanner

RABIN KARP

pattern: 71344

71344 % 997 = 557

i	6	5	3	2	1	7	1	3	4	4	5	6
0	6	5	3	2	1	% 997 = 516 X						
1		5	3	2	1	7	% 997 = 376 X					
2			3	2	1	7	1	997 = 267 X				
3				2	1	7	1	3	% 997 = 776 X			
4					1	7	1	3	4	% 997 = 185 X		
5						7	1	3	4	4	% 997 = 557	

↳ return i=5

Scanned with CamScanner

KMP (KNUTH-MORIS-PRATT) — DFA

pattern = ANANAS

	A	N	A	N	A	S
A	1	1	3	1	5	1
N	0	2	0	4	0	4
S	0	0	0	0	0	6

1° A

A — 1

N — 0

S — 0

2° AN

AA — 1

AN — 2

AS — 0

3° ANA

ANA — 3

ANN — 0

ANS — 0

4° ANAN

ANAA — 1

ANAN — 4

ANAS — 0

5° ANANA

ANANA — 5

ANANN — 0

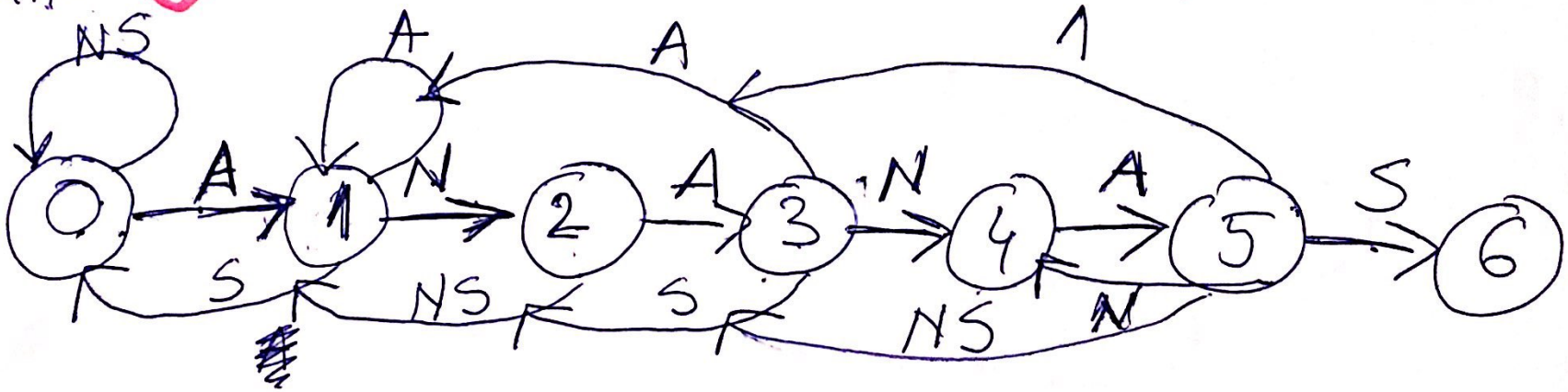
ANANS — 0

6° ANANAS

ANANAA — 1

ANANAN — 4

ANANAS — 6



Scanned with CamScanner

(*) BANANA

	B	A	N	A	N	A
A	0	2	0	4	0	6
B	1	1	1	1	1	1
N	0	0	3	0	5	0

1° B
A - 0
B - 1
N - 0

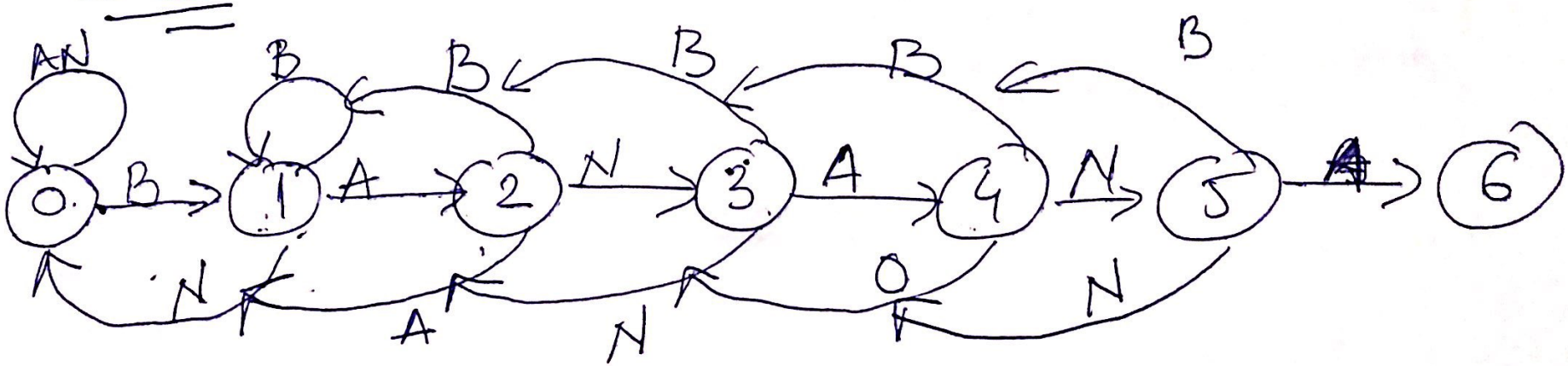
2° BA
BA - 2
BB - 1
BN - 0

3° BAN
BAA - 0
BAB - 1
BAN - 3

4° BANA
BANA - 4
BANB - 1
BANN - 0

5° BANAN
BANAA - 0
BANAB - 1
BANAN - 5

6° BANANA
BANANA - 6
BANANB - 1
BANANN - 0



Scanned with CamScanner